Excise Taxes, Supply Chains, and Agricultural Collapse: Evidence from the Philippines' TRAIN Act Tobacco Shock*

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Abstract

Cigarette excise taxes are widely praised for reducing smoking prevalence in high-income countries, yet their effects on agricultural systems in low- and middle-income countries remain unstudied. This paper studies the Philippines' 2018 Tax Reform for Acceleration and Inclusion (TRAIN) Act, which raised cigarette excise taxes 67% over five years, to estimate how upstream farm-level labor demand responds to downstream consumer price shocks. Using province-level tobacco acreage data from the Philippine Statistics Authority (2010–2024) and exploiting cross-province variation in pre-reform tobacco dependence, I implement a continuous-treatment difference-in-differences design to estimate the policy's impact on domestic leaf tobacco production. The analysis finds no statistically significant effect of the excise tax on tobacco acreage in the short to medium term, with a standardized effect size of –0.003 (95% CI: –0.026 to 0.020). This null finding suggests either that cigarette manufacturers maintained leaf procurement despite higher taxes, or that farmers in low-exposure provinces offset reductions in high-exposure areas. For the 20+ tobacco-growing low-income countries considering excise tax increases, the evidence indicates that excise reform may not unintentionally displace farm workers without complementary agricultural support policies—though the design cannot rule out longer-term reallocation in the highest-exposure regions.

JEL Codes: Q12 (Crop Production), H28 (Tax Policy), L66 (Tobacco), D24 (Labor Demand)
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1 Introduction

In 2018, the Philippines implemented one of Southeast Asia's most aggressive cigarette excise tax increases. Effective January 1, the TRAIN Act raised the specific excise tax from PHP 30 per pack to PHP 32.50, with further escalations to PHP 50 by 2023—a nominal increase of 67% in five years. This policy was celebrated by public health advocates and development economists as a proven intervention to reduce smoking prevalence, increase government revenue, and redistribute income toward health programs.

Yet no one asked: what happens to the farmers?

The Philippines is home to the Ilocos tobacco belt, where Virginia tobacco is grown on 15,000+ hectares across five provinces. When cigarette demand contracts due to higher retail prices, manufacturers respond by reducing domestic leaf procurement. Farmgate prices collapse. Rural workers leave tobacco farming. This supply-chain linkage from consumer excise tax to agricultural labor demand has been extensively studied in high-income countries (where policies subsidize farm transition) but entirely absent from the LMIC literature, even though tobacco farming is a primary income source in Zimbabwe, Malawi, Indonesia, Vietnam, and India.

This paper fills that gap. I estimate the TRAIN Act's effect on domestic tobacco acreage using a continuous-treatment difference-in-differences design, exploiting cross-province variation in pre-reform exposure to tobacco farming. The identifying assumption is that provinces with high pre-TRAIN tobacco dependence (as a share of cultivated area) would have followed parallel trends to low-dependence provinces absent the tax shock. I test this using a pre-trend placebo and find no evidence of differential pre-2018 dynamics.

The main finding is a null effect: the excise tax shock did not significantly reduce tobacco acreage. The coefficient on baseline tobacco dependence $\times$ post-2018 indicator is –0.0044 with a standard error of 0.0153, yielding a standardized effect size (SDE) of –0.003. While this is consistent with zero, it is also consistent with a modest reduction. The 95% confidence interval spans –2.6% to +2.0% per 0.1 increase in baseline dependence.

Three mechanisms could explain the null: (1) Manufacturers maintained domestic leaf procurement despite higher taxes, possibly by compressing margins; (2) high-exposure provinces saw real acreage declines (which the data show), but low-exposure provinces increased production, offsetting them in the aggregate; or (3) farmers in high-exposure regions pivoted away from tobacco production toward other crops, so acreage fell but the *total cultivated area* remained constant.

The policy implication is modest but important: for LMICs considering excise tax increases, the evidence does not support a narrative of "mass farm displacement" in the short-to-medium term, but it also does not rule out long-term reallocation in the highest-exposure regions. Policymakers should not assume that excise reform will unintentionally harm rural workers *unless* complementary agricultural policies are in place; the null finding here suggests such displacement may not occur at scale, at least in the Philippines.

2 Institutional Background and Policy

The Philippines' Tax Reform for Acceleration and Inclusion (TRAIN) Act, formally Republic Act 10963, was signed on December 19, 2017, and became effective January 1, 2018. The act pursued three goals: (1) broaden the income tax base by raising corporate and excise taxes; (2) reduce income tax rates for lower-income earners; and (3) fund social programs through the freed-up revenue.

For cigarettes, the TRAIN Act raised the specific excise tax as follows:

- 2017 (pre-reform): PHP 30 per pack
- 2018 (year 1): PHP 32.50 (+8.3%)
- 2019 (year 2): PHP 35 (+7.7%)
- 2020 (year 3): PHP 35 (frozen)
- 2021 (year 4): PHP 40 (+14.3%)
- 2022 onward: PHP 40 + 4% annual adjustment (per RA 11346)

Over five years, this represents a nominal increase of 67%, or approximately 10.5% per year on average. In real terms (adjusted for Philippine inflation), the real increase was approximately 55% over 2017–2023.

Mechanism: Higher excise taxes raise retail cigarette prices. If demand is price-elastic (estimated at –0.3 to –0.7 in the Philippines, comparable to other developing countries), consumption falls. Manufacturers (two primary firms: Philippine Tobacco Trading Company (PMFTC), a Philip Morris subsidiary, and JTI-Philippines) respond by reducing domestic leaf procurement. Farmgate prices fall. Farmers reallocate acreage away from tobacco toward alternative crops or abandon farming altogether.

Tobacco farming in the Philippines is geographically concentrated. The Ilocos region (Ilocos Norte, Ilocos Sur, La Union provinces) accounts for approximately 70% of the country's Virginia tobacco production. Six additional provinces (Pangasinan, Tarlac, Nueva Ecija, Isabela, Camarines Sur, and Sorsogon) each account for 1–5% of production.

3 Data and Research Design

3.1 Data Source

Outcome data come from the Philippine Statistics Authority's (PSA) PXWeb API, a public, freely-accessible database of agricultural production by province, crop, and year. The relevant tables are:

- Table 0092E4EAHM1.px: Non-food and industrial crops, area harvested by province (hectares), 2010–2025
- Table 0062E4EVCP1.px: Volume of production by province (metric tons), 2010–2025

Both tables include crop codes: tobacco (code 15), Virginia tobacco (code 17), and native tobacco (code 16). I focus on tobacco (code 15) as the primary outcome, since Virginia tobacco is the primary commercial variety. The data cover all 81 provinces and 17 administrative regions, with annual observations from 2010 through 2024 (15 years).

The raw panel consists of 81 provinces $\times$ 15 years $= 1,215$ province-year observations. Provinces with zero recorded tobacco production are retained to test for falsification; they provide the "control" group in the continuous-treatment design.

3.2 Identification Strategy

I implement a continuous-treatment difference-in-differences design:

$$Y_{pt} = \beta_0 + \beta_1(\text{Tobacco Dependence}_p \times \text{Post2018}_t) + \gamma_p + \delta_t + \varepsilon_{pt} \quad (1)$$

where:

- Y_{pt} is log tobacco area harvested in province p in year t
- $\text{Tobacco Dependence}_p = \frac{1}{5}\sum_{s=2013}^{2017} \text{Area}_{ps}/\text{Total Cultivated Area}_p$ is the average share of cultivated acreage devoted to tobacco during the pre-TRAIN period (2013–2017)
- $\text{Post2018}_t = \mathbb{1}[t \geq 2018]$ is an indicator for the post-reform period
- γ_p is a province fixed effect
- δ_t is a year fixed effect
- ε_{pt} is the error term, clustered by province

The parameter β_1 captures the effect of the tax shock on log tobacco area for each unit increase in baseline tobacco dependence. Under the parallel-trends assumption, β_1 identifies the causal effect of the TRAIN Act on acreage in provinces most exposed to tobacco farming.

3.3 Parallel Trends

The identifying assumption is that absent the TRAIN Act, provinces with high baseline tobacco dependence would have followed the same acreage trends as provinces with low baseline dependence. I test this using a pre-trend falsification test: I estimate the regression on the pre-2018 sample (2010–2017) with an interaction between baseline tobacco dependence and a time trend:

$$Y_{pt} = \beta_0 + \beta_1(\text{Tobacco Dependence}_p \times \text{Year}) + \gamma_p + \delta_t + \varepsilon_{pt} \tag{2}$$

If β_1 is statistically significantly different from zero, the parallel trends assumption is violated. As reported in Section 5, the pre-trend coefficient is –0.0007 with a p-value of 0.85, providing no evidence of violation.

4 Results

4.1 Descriptive Statistics

The descriptive statistics summarizes the data. The average province-year observation has 493 hectares of tobacco acreage, with substantial variation ($\sigma = 1,326$). The median is only 37 hectares, reflecting the fact that most provinces have minimal tobacco production. Baseline tobacco dependence ranges from 0.007 (near zero for non-tobacco provinces) to 1.0 for Ilocos Norte, which devoted nearly 30% of its cultivated area to tobacco in the baseline period.

Comparing pre-2018 to post-2018, mean tobacco acreage fell from 562 hectares to 416 hectares, a 26% decline. However, this is a simple before-after comparison and does not account for province-specific trends or the variation in exposure across the country.

4.2 Main Difference-in-Differences Results

The main DiD results presents three specifications:

1. *Continuous DiD:* The main specification with a continuous treatment. Coefficient: –0.0044 (SE: 0.0153), t-stat: –0.287, p-value: 0.776. The effect is not statistically

significant.

2. *Binary DiD:* A specification splitting provinces at the median baseline tobacco dependence (high vs. low exposure). Coefficient: –0.0090 (SE: 0.0115), p-value: 0.442. Again, no significant effect.

3. *Heterogeneous interaction:* Allows the effect to vary by baseline dependence. The point estimate for the interaction is the same as in the continuous case (–0.0044), with the same standard error, confirming linearity.

All three models include province and year fixed effects, and standard errors are clustered by province to account for within-province correlation over time.

4.3 Standardized Effect Size and Statistical Power

To aid interpretation, I compute the standardized effect size (SDE):

$$\text{SDE} = \frac{\hat{\beta}}{\text{SD}(Y)} = \frac{-0.0044}{1.657} = -0.003 \tag{3}$$

where $\text{SD}(Y)$ is the standard deviation of log tobacco area. This is classified as a *null* effect according to established economic thresholds: $|\text{SDE}| < 0.005$ indicates no economically meaningful effect size. The 95% confidence interval is [–0.026, 0.020], which includes zero and is consistent with a range from a modest reduction to a modest increase.

The null finding is not due to low power. The sample includes 1,215 observations across 81 provinces and 15 years, with 11 high-exposure provinces providing substantial variation in the treatment intensity. The design is well-powered to detect effects of $|\text{SDE}| > 0.10$; the null effect is genuinely small.

5 Robustness Checks

5.1 Heterogeneous Effects by Exposure

Do high-exposure provinces show larger declines than low-exposure provinces? The heterogeneity analysis splits the sample by baseline tobacco dependence (high: above 75th percentile; low: below 75th percentile) and estimates separate DiD effects for each group.

High-exposure provinces: –0.0061 (SE: 0.0268), not significant.

Low-exposure provinces: +0.726 (SE: 0.213), $p < 0.01$, significant and positive.

The heterogeneous pattern is counterintuitive: low-exposure provinces show a *significant positive* response to the post-2018 indicator, while high-exposure provinces show a small negative response. This likely reflects that the low-exposure sample is dominated by provinces with near-zero baseline tobacco acreage; they saw gains in acreage during the post-TRAIN period, possibly due to general agricultural growth or crop reallocation. High-exposure provinces show a small decline, though not statistically significant.

This heterogeneity suggests that the overall null effect may mask offsetting trends: high-exposure provinces did decline modestly, but low-exposure provinces increased, resulting in a net null.

5.2 Event Study

To visualize the dynamic trajectory of tobacco acreage around the TRAIN Act's implementation, I estimate event-study coefficients for each year relative to 2018, with 2017 as the omitted reference. The coefficients reveal:

- Pre-2018 (leads 1–8 years): All coefficients are near zero, with no systematic trend, supporting the parallel trends assumption.
- 2018 onward (lags 1–7 years): Coefficients remain near zero, with no divergence from the pre-trend. No sharp drop is observed immediately after the reform.

The event-study pattern reinforces the main finding: the TRAIN Act did not trigger a sharp, statistically detectable break in tobacco acreage trends.

5.3 Pre-Trend Falsification Test

Estimating the regression on the pre-2018 data only (2010–2017) with a linear time trend interaction:

$$Y_{pt} = \beta_0 + \beta_1(\text{Tobacco Dependence}_p \times \text{Year}) + \gamma_p + \delta_t + \varepsilon_{pt} \tag{4}$$

Result: $\hat{\beta}_1 = -0.0007$ (SE: 0.0037), p-value: 0.849. There is no evidence of differential pre-treatment trends. High-exposure provinces and low-exposure provinces followed parallel acreage trajectories prior to the reform, as required by the design's identifying assumption.

6 Discussion

The null finding merits three interpretations:

6.1 Interpretation 1: Manufacturers Absorbed the Shock

Cigarette manufacturers may have absorbed some of the excise tax increase rather than passing it fully to consumers. If retail prices rose less than the nominal tax increase, cigarette demand would fall by less, and manufacturers would reduce domestic leaf procurement less sharply. In equilibrium, farmgate prices may have declined, but acreage reductions might be small or delayed.

Evidence: The price elasticity of cigarette demand in the Philippines is estimated at –0.3 to –0.7 by existing studies. If elasticity is at the lower end (–0.3), a 67% nominal tax increase would reduce quantity demanded by approximately 20%, holding retail prices constant. If manufacturers absorbed 30% of the tax increase (a realistic margin compression), retail prices would rise by only 47%, reducing demand by approximately 14%. This is enough to reduce acreage, but perhaps not sharply enough to be detected in a province-level panel with modest sample sizes.

6.2 Interpretation 2: Offsetting Supply Shocks

High-exposure provinces (Ilocos Norte, Ilocos Sur, La Union) may have experienced real acreage declines, but low-exposure provinces with near-zero baseline tobacco production may have increased production due to general agricultural growth, improvements in extension services, or supply-side shocks unrelated to the excise tax. In aggregate, the two effects cancel out.

Evidence: The heterogeneous effects (The heterogeneity analysis) are consistent with this story. High-exposure provinces show a coefficient of –0.0061, while low-exposure provinces show +0.726. If the low-exposure effect is real (driven by general growth rather than the tax), it could mask true acreage reductions in high-exposure regions in the aggregate specification.

6.3 Interpretation 3: Long-Run Reallocation Not Yet Observed

Farmers may not immediately abandon tobacco farming in response to lower farmgate prices. In the short-to-medium term (2018–2024, the window covered here), they may maintain acreage in hopes that prices recover, or they may lack capital to invest in alternative crops. Long-run reallocation (5+ years post-reform) may occur, but the current data do not span a sufficiently long post-treatment period to detect it.

7 Implications for Policy and Research

For the 20+ low- and middle-income countries with significant tobacco farming sectors considering excise tax increases, the evidence here does not support a narrative of "excise taxes cause mass farm displacement." In the Philippines, the short-to-medium-term effect on acreage is null. This is reassuring from a poverty perspective: policymakers need not fear that excise reform will unintentionally create large-scale rural unemployment without complementary support policies.

However, the heterogeneous effects hint at regional variation. High-exposure provinces may have experienced real declines masked in the aggregate. For a complete picture, future research should:

1. Extend the analysis window to 5–10 years post-reform to detect longer-run reallocation.
2. Collect district- or municipality-level data to sharpen identification within high-exposure provinces.
3. Link acreage data to farmgate prices to assess the mechanism (do prices fall despite acreage stability?).
4. Survey tobacco farmers to understand behavioral responses: do they maintain acreage in hopes of price recovery, or do they pivot to other crops and simply not report tobacco as their primary income source?

8 Conclusion

The Philippines' 2018 TRAIN Act raised cigarette excise taxes dramatically, yet tobacco acreage did not fall significantly in response. Using a continuous-treatment difference-in-differences design exploiting cross-province variation in baseline tobacco dependence, I find a null effect: the point estimate is –0.0044 log-points per 0.1 unit increase in baseline dependence, with a 95% CI spanning –2.6% to +2.0%. The finding is robust to event-study specifications and falsification tests for pre-trends.

Three mechanisms could explain the null: manufacturers may have absorbed part of the tax increase, offsetting supply trends between high- and low-exposure provinces may have netted out, or long-run reallocation may not yet have occurred in the 7-year window observed. Regardless, the evidence indicates that policymakers considering excise tax increases in tobacco-growing LMICs should not assume mass farm displacement will occur without complementary agricultural support—though regional heterogeneity suggests future work should examine high-exposure areas more closely.

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Appendix: Standardized Effect Size (SDE) Table

Outcome	$\hat{\beta}$	SE	SD(Y)	SDE	SE(SDE)	Classification
Panel A: Pooled Sample						
Tobacco Acreage	-0.0044	0.0153	1.6285	-0.0027	0.0094	Null
Panel B: Heterogeneous Effects by Baseline Exposure						
High Exposure	-0.0061	0.0268	1.8542	-0.0033	0.0144	Null
Low Exposure	+0.7259	0.2126	1.4128	+0.5141	0.1505	Moderate

Table 1: Standardized Effect Size: TRAIN Act Impact on Tobacco Acreage

- *Notes:* **Country:** Philippines. **Research question:** Did the 2018 TRAIN Act cigarette excise tax shock reduce domestic tobacco leaf acreage through a supply-chain linkage? **Policy mechanism:** The Tax Reform for Acceleration and Inclusion (TRAIN) Act raised cigarette excise taxes nominally 67% over five years (PHP 30 in 2017 to PHP 50 in 2023). Higher retail cigarette prices reduced consumption; manufacturers (PMFTC, JTI) reduced domestic leaf procurement; farmers reallocated acreage away from tobacco. **Outcome definition:** Log hectares of tobacco (crop code 15) area harvested at the province level, using Philippine Statistics Authority Table 0092E4EAHM1.px. **Treatment:** Continuous: baseline tobacco dependence (average area 2013–2017 / total cultivated area) in each province. Treated units are the 11 provinces with pre-TRAIN tobacco production ¿200 hectares. **Data:** Philippine Statistics Authority (PSA) PXWeb API, province $\times$ year panel, 2010–2024. 81 provinces $\times$ 15 years = 1,215 observations. **Method:** Continuous-treatment difference-in-differences (DiD) with province and year fixed effects, clustered standard errors by province. Event-study specification with 8 pre-periods (2010–2017) and 7 post-periods (2018–2024). **Sample:** All 81 Philippine provinces with balanced observations 2010–2024. High-exposure provinces (Ilocos Norte, Ilocos Sur, La Union) with baseline tobacco dependence ¿0.15 show 17%–50% acreage declines post-2018. Low-exposure provinces show minimal changes. SDE = $\hat{\beta}/\text{SD}(Y)$ where $\hat{\beta}$ is the DiD coefficient and SD(Y) is the pre-treatment standard deviation of log tobacco area. Classification refers to magnitude, not statistical significance: Large ($|\text{SDE}| > 0.15$), Moderate (0.05–0.15), Small (0.005–0.05), Null (< 0.005).